

KRAKENES FYR, Norway

WMO: 012030

Lat: 62.0342N

Lon: 4.9862E

Elev: 135

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.7	28.9	4.2	6.9	43.5	9.4	9.0	36.2	59.5	43.6	55.3	42.9	11.1	80	0.911

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	70.4	59.6	67.4	58.3	64.8	57.4	62.3	68.0	60.7	65.3	59.2	63.6	11.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.2	78.2	63.4	58.5	73.5	62.5	56.9	69.3	61.4	28.0	68.5	26.9	65.4	25.9	63.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 56.0	50.2	44.9	25.4	76.2	3.0	4.4	23.3	79.4	21.6	82.0	19.9	84.5	17.7	87.7		
(5)			21.7	64.6	3.4	2.0	19.2	66.1	17.3	67.2	15.4	68.3	12.9	69.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	38.5	38.2	39.0	43.3	47.6	52.2	57.2	58.1	55.1	48.5	43.3	40.2		
	DBStd	8.58	4.34	4.45	4.33	5.04	5.56	4.30	4.97	4.18	4.45	5.13	5.24	5.11		
	HDD50	1981	355	331	342	211	114	22	1	1	7	87	207	303		
	HDD65	6651	820	751	806	652	540	386	250	218	299	510	651	768		
	CDD50	818	0	0	0	9	40	86	224	253	158	42	6	0		
	CDD65	16	0	0	0	0	0	1	8	6	1	0	0	0		
	CDH74	23	0	0	0	0	0	0	19	4	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	19.3	21.8	20.8	20.2	18.9	17.0	16.6	14.9	15.7	20.2	20.9	21.0	23.1		
	PrecAvg	73.1	8.0	6.7	6.2	4.3	3.3	3.9	4.4	5.6	6.5	7.8	7.8	8.8		
(15) Precipitation	PrecMax	96.5	14.3	13.5	12.6	8.4	5.1	8.7	6.9	10.0	11.4	14.1	14.4	15.7		
	PrecMin	51.8	1.5	0.6	1.3	1.1	0.8	0.5	1.0	2.2	2.2	2.4	1.8	2.2		
	PrecStd	11.3	3.4	3.1	2.7	1.5	1.1	1.8	1.4	1.8	2.9	2.7	3.3	3.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.7	51.3	49.8	61.1	68.5	70.0	76.4	73.0	70.0	66.2	57.6	52.7		
		MCWB	39.7	43.7	43.5	47.5	54.7	55.3	61.8	62.6	58.8	55.9	49.4	45.1		
	2%	DB	46.7	47.1	47.4	56.6	63.5	64.7	71.1	69.2	66.3	60.3	53.7	49.8		
		MCWB	43.3	42.4	42.1	46.7	52.8	54.2	61.1	58.8	56.4	50.6	47.4	45.1		
	5%	DB	45.5	45.4	45.8	53.4	58.9	60.3	67.3	66.3	63.5	57.1	51.8	48.2		
		MCWB	42.3	41.5	41.8	45.4	50.6	53.3	59.7	58.9	55.3	50.7	46.7	45.0		
	10%	DB	44.5	44.2	44.4	50.3	55.1	57.5	64.3	63.7	61.2	55.0	50.2	47.0		
		MCWB	41.2	40.7	40.8	43.9	49.4	51.9	58.9	58.2	54.8	49.6	45.9	43.5		
	0.4%	WB	45.9	46.2	46.2	51.6	59.1	58.5	65.3	64.3	60.9	56.2	51.1	48.3		
		MCDB	46.8	48.1	47.2	56.1	65.8	65.1	69.2	68.0	66.1	61.5	54.4	50.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.7	44.5	44.4	48.8	54.1	56.4	62.9	61.9	58.8	53.8	49.6	46.8		
		MCDB	45.8	46.0	46.1	53.9	60.4	62.3	69.0	66.5	63.9	57.7	52.2	48.5		
	5%	WB	43.4	43.1	43.1	45.9	51.8	54.8	61.4	60.4	57.1	52.3	47.9	45.5		
		MCDB	45.1	44.8	45.1	51.6	57.4	58.9	67.2	64.0	61.3	56.1	50.6	47.4		
	10%	WB	41.6	41.4	41.5	44.3	49.6	53.3	59.5	59.2	55.7	50.5	46.1	43.7		
		MCDB	43.8	43.7	43.8	49.4	54.8	56.4	64.8	62.7	59.6	54.1	49.4	46.1		
	5% DB	MDBR	5.2	5.0	5.5	6.5	6.7	6.2	6.4	5.9	6.0	5.7	5.5	5.1		
		MCDBR	6.0	6.3	7.1	10.9	12.3	10.8	11.5	9.5	8.9	8.2	6.8	6.5		
		MCWBR	5.5	5.4	5.6	5.1	5.7	6.0	5.6	5.2	5.0	6.0	5.2	6.0		
		MCWBR	5.5	6.1	6.2	9.8	11.8	9.3	10.8	7.7	7.6	8.0	6.5	6.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.1	6.1	6.0	5.8	6.3	5.9	5.7	5.3	5.5	7.0	6.6	6.7		
	taub		0.288	0.329	0.349	0.372	0.362	0.358	0.371	0.365	0.353	0.328	0.289	0.366		
		taud	2.083	2.223	2.301	2.320	2.416	2.466	2.455	2.470	2.471	2.372	2.151	2.385		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		117	195	239	258	272	275	268	259	239	197	117	66		
		Edh at Noon	12	21	27	33	32	31	31	28	23	18	11	7		
	RadAvg		52	197	534	1065	1536	1615	1456	1100	646	298	81	30		
(44) All-Sky Solar Radiation	RadStd		8	36	86	116	105	180	121	117	92	48	17	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	+1.22	N/A	N/A	N/A	N/A	-166	-243	N/A	N/A

Nomenclature: See separate page